

Analysing Smart City Bike Sharing Data

1. Overview and Objective

This project involves cleaning, transforming, and analysing raw data and creating an interactive Power BI dashboard to derive meaningful business insights.

The main objective is to demonstrate data pre-processing techniques using an interactive Power BI dashboard visualization to make informed decisions.

2. Problem Statement

- To analyse the performance of bike sharing station across cities to identify top-performing cities.
- To study patterns across cities and years for planning.

3. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Name	String (Text)	Name of the location
Address	String (Text)	Name of the location
Position	Numeric (Integer)	Latitude and Longitude co-ordinates
Banking	Boolean	Indicates whether card payment is accepted
Bonus	Boolean	Indicates whether bonus is offered
Status	Boolean	Denotes whether station is open or closed
City	Categorical	Name of the city
Bike stands	Numeric (Integer)	Number of bike stands
Available bike stands	Numeric (Integer)	Number of Available bike stands
Available Bikes	Numeric (Integer)	Number of Available Bikes
Date	Numeric (Integer)	Last updated Date
Time	Numeric (Integer)	Last updated time

4. Tools & Technologies

- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

- **Data Cleaning & Transformation:** Handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- Split address column to separate address from number.

- Formatted the Contract name column to proper case and renamed the column to City.

	A6C	Contract Name	123 Bike Stands	123 Available Bike Stands	123 Available Bikes	Last Modified
1		Maribor	1	0	0	
2		Marseille	9	0	0	
3		Marseille	15	0	0	
4		Marseille	15	15	0	
5		Nantes	0	0	0	
6		Marseille	10	0	0	
7		Marseille	24	24	0	
8		Marseille	20	13	7	
9		Marseille	10	0	0	
10		Marseille	20	2	0	
11		Marseille	10	10	0	
12		Marseille	14	0	0	
13		Marseille	15	0	0	
14		Marseille	10	0	0	
15		Marseille	8	0	0	
16		Marseille	32	0	0	
17		Marseille	16	0	0	
18		Marseille	17	0	0	
19		Jcdecauxbike	1	0	1	
20		Marseille	10	0	0	
21						

= Table.TransformColumns("#Filtered Rows2",{{"Contract Name", Text.Proper, type text}})

Query Settings

PROPERTIES

Name
Feuil1

All Properties

APPLIED STEPS

- Filtered Rows
- Removed Duplicates
- Renamed Column Number
- Split Column by Delimiter ⚙️
- Changed Type1
- Split Column by Delimiter1 ⚙️
- Changed Type2
- Filtered Rows1
- Merged Columns ⚙️
- Trimmed Text
- Cleaned Text
- Filtered Rows2
- Capitalized Each Word
- Split Column by Delimiter2 ⚙️
- Changed Type3
- Renamed Columns

- Split Last update column in to three (Date, Time and Time zone).

Query Editor: `= Table.SplitColumn(Table.TransformColumnTypes("#Capitalized Each Word", {"Last Update", type text}),`

	Available Bike Stands	Available Bikes	APC Last Update.1	APC Last Update.2	APC Last Update.3
1	0	0	02-06-2022	15:14:38	+05:30
2	0	0	06-04-2022	12:54:35	+05:30
3	0	0	27-09-2022	10:57:48	+05:30
4	15	0	08-06-2022	11:50:13	+05:30
5	0	0	23-05-2022	20:11:50	+05:30
6	0	0	13-09-2022	11:28:01	+05:30
7	24	0	06-09-2022	18:02:17	+05:30
8	13	7	14-09-2022	13:27:09	+05:30
9	0	0	12-09-2022	11:18:28	+05:30
10	2	0	03-10-2022	11:34:01	+05:30
11	10	0	13-09-2022	15:41:24	+05:30
12	0	0	26-09-2022	16:44:57	+05:30
13	0	0	03-10-2022	12:02:43	+05:30
14	0	0	10-10-2022	12:40:37	+05:30
15	0	0	26-09-2022	11:29:34	+05:30
16	0	0	17-10-2022	11:36:42	+05:30
17	0	0	24-10-2022	13:08:08	+05:30
18	0	0	04-10-2022	11:49:03	+05:30
19	0	1	30-08-2022	21:00:02	+05:30
20	0	0	08-11-2022	14:53:04	+05:30
21	-	-	-	-	-

Query Settings: **PROPERTIES** Name: Feuil1. **APPLIED STEPS**: Filtered Rows, Removed Duplicates, Renamed Column Number, Split Column by Delimiter, Changed Type1, Split Column by Delimiter1, Changed Type2, Filtered Rows1, Merged Columns, Trimmed Text, Cleaned Text, Filtered Rows2, Capitalized Each Word, Split Column by Delimiter2, Changed Type3, Renamed Columns.

- Created the following calculated columns:

1. Stand Availability rate (%):

1 Stand availability rate (%) = DIVIDE('Fact'[Available Bike Stands], 'Fact'[Bike Stands])*100

nds

Available Bikes

Last updated Date

Last updated Time

Time Zone

Occupied stands

Bike availability rate (%)

Stand availability rate (%)

1

2

13

9

1

18

20

8

0

7

1

19

18

1

0

6

13

8

11

2

12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

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12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

12 November 2025

2.49.15 PM

2.49.03 PM

2.49.08 PM

2.49.54 PM

2.50.37 PM

2.49.18 PM

2.51.22 PM

2.48.30 PM

2.50.54 PM

2.51.12 PM

2.50.26 PM

2.51.48 PM

2.49.04 PM

2.49.15 PM

2.49.31 PM

2.48.55 PM

2.53.34 PM

2.52.08 PM

2.49.37 PM

2.48.29 PM

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

+05:30

2

2

13

18

1

19

20

8

0

7

3

20

18

1

0

8

13

10

11

2

5

10

65

45

5

90

100

40

0

35

5

95

0

100

30

65

40

55

10

90

90

35

10

95

5

0

60

100

65

85

0

10

95

100

60

35

50

45

90

Data

Search

Dim_City

City

Fact

Address

Available Bike Stands

Available Bikes

Banking

Bike availability rate (%)

Bike Stands

Bonus

City

Last updated Date

Last updated Time

Number

Occupied stands

Position

Stand availability rate (%)

2. Bike Availability rate (%):

1 Bike availability rate (%) = DIVIDE('Fact'[Available Bikes], 'Fact'[Bike Stands])*100

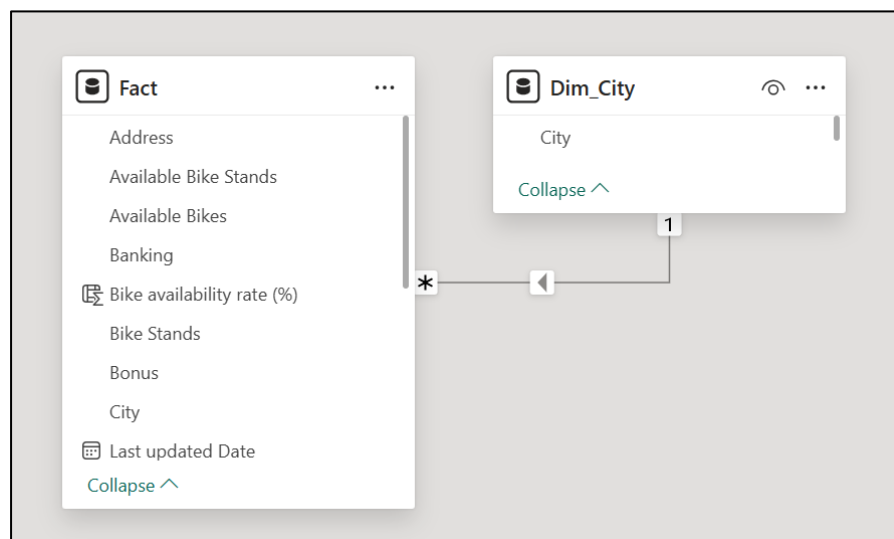
nds	Available Bikes	Last updated Date	Last updated Time	Time Zone	Occupied stands	Bike availability rate (%)	Stand availability rate (%)
	1	12 November 2025	2.49.15 PM	+05:30	2	5	90
	2	12 November 2025	2.49.03 PM	+05:30	2	10	90
	13	12 November 2025	2.49.08 PM	+05:30	13	65	35
	9	12 November 2025	2.49.54 PM	+05:30	18	45	10
	1	12 November 2025	2.50.37 PM	+05:30	1	5	95
	18	12 November 2025	2.49.18 PM	+05:30	19	90	5
	20	12 November 2025	2.51.22 PM	+05:30	20	100	0
	8	12 November 2025	2.48.30 PM	+05:30	8	40	60
	0	12 November 2025	2.50.54 PM	+05:30	0	0	100
	7	12 November 2025	2.51.12 PM	+05:30	7	35	65
	1	12 November 2025	2.50.26 PM	+05:30	3	5	85
	19	12 November 2025	2.51.48 PM	+05:30	20	95	0
	18	12 November 2025	2.49.04 PM	+05:30	18	90	10
	1	12 November 2025	2.49.15 PM	+05:30	1	5	95
	0	12 November 2025	2.49.31 PM	+05:30	0	0	100
	6	12 November 2025	2.48.55 PM	+05:30	8	30	60
	13	12 November 2025	2.53.34 PM	+05:30	13	65	35
	8	12 November 2025	2.52.08 PM	+05:30	10	40	50
	11	12 November 2025	2.49.37 PM	+05:30	11	55	45
	2	12 November 2025	2.48.29 PM	2.49.37 PM	2	10	90

3. Stand availability status:

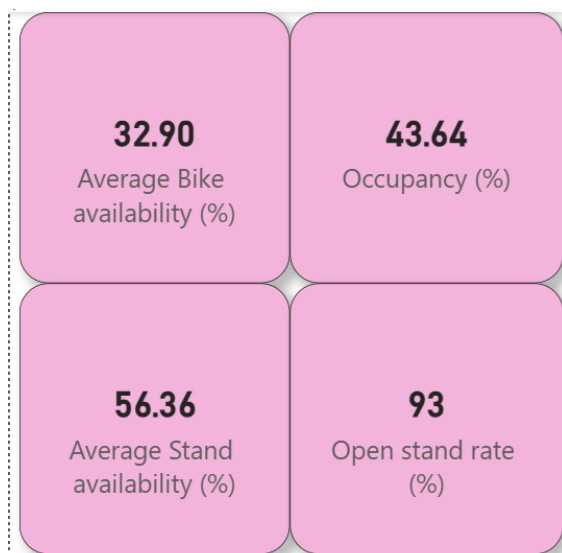
1 Stand availability = IF('Fact'[Stand availability rate (%)>70, "High", IF('Fact'[Stand availability rate (%)>30,"Medium","Low"))							Data
Last updated Date	Last updated Time	Time Zone	Occupied stands	Bike availability rate (%)	Stand availability rate (%)	Stand availability	Search
12 November 2025	2.49.15 PM	+05:30	2	5	90	High	Available Bike Stands
12 November 2025	2.49.03 PM	+05:30	2	10	90	High	Available Bikes
12 November 2025	2.49.08 PM	+05:30	13	65	35	Medium	Banking
12 November 2025	2.49.54 PM	+05:30	18	45	10	Low	Bike availability rate (%)
12 November 2025	2.50.37 PM	+05:30	1	5	95	High	Bike Stands
12 November 2025	2.49.18 PM	+05:30	19	90	5	Low	Bonus
12 November 2025	2.51.22 PM	+05:30	20	100	0	Low	City
12 November 2025	2.48.30 PM	+05:30	8	40	60	Medium	> Last updated Date
12 November 2025	2.50.54 PM	+05:30	0	0	100	High	Last updated Time
12 November 2025	2.51.12 PM	+05:30	7	35	65	Medium	Number
12 November 2025	2.50.26 PM	+05:30	3	5	85	High	Occupied stands
12 November 2025	2.51.48 PM	+05:30	20	95	0	Low	Position
12 November 2025	2.49.04 PM	+05:30	18	90	10	Low	Stand availability rate (%)
12 November 2025	2.49.15 PM	+05:30	1	5	95	High	Stand availability
12 November 2025	2.49.31 PM	+05:30	0	0	100	High	Station
12 November 2025	2.48.55 PM	+05:30	8	30	60	Medium	Status
12 November 2025	2.53.34 PM	+05:30	13	65	35	Medium	Time Zone
12 November 2025	2.52.08 PM	+05:30	10	40	50	Medium	
12 November 2025	2.49.37 PM	+05:30	11	55	45	Medium	
12 November 2025	2.48.29 PM	+05:30	2	10	90	High	

6. Data Modelling and DAX (Power BI)

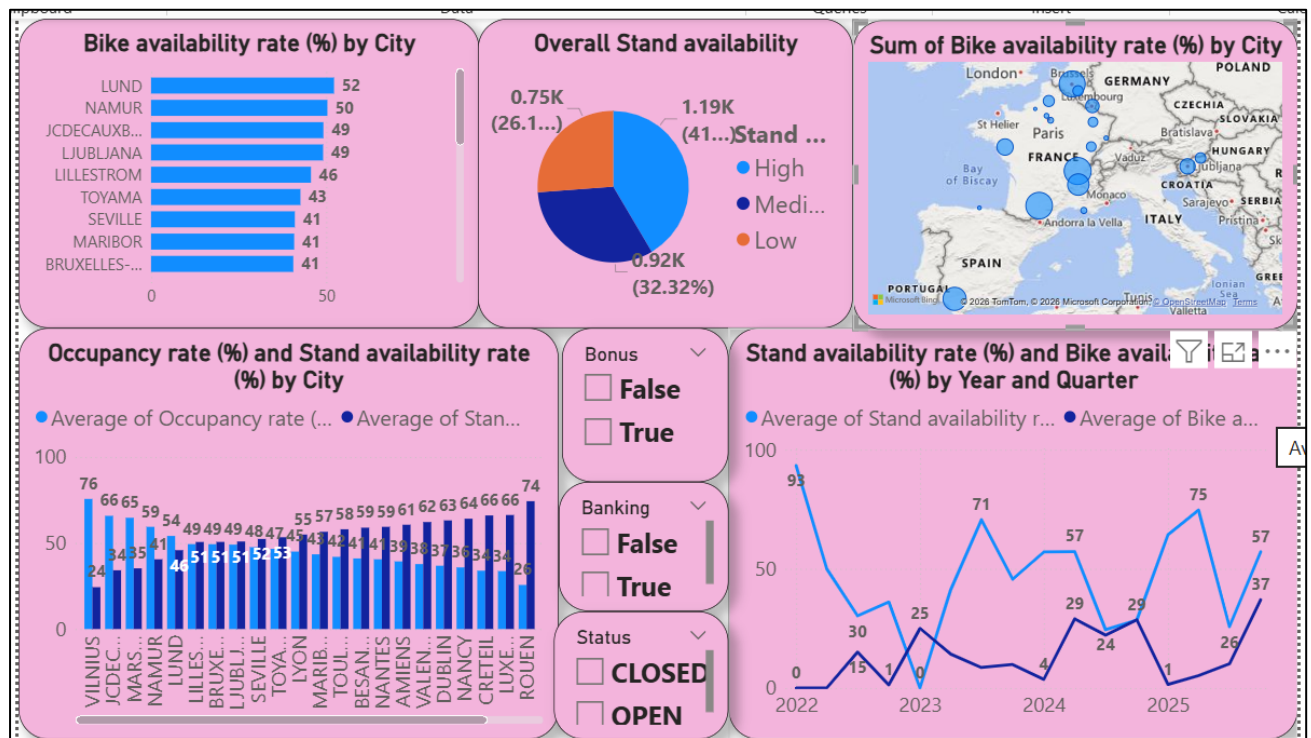
- **Data Model:** Established relationships between tables and defined cardinality.



- **DAX Measures:** Implemented DAX formulas for key metrics, such as Average Bike availability rate (%), Occupancy (%), Average Stand availability rate (%), and Open Stand rate (%).

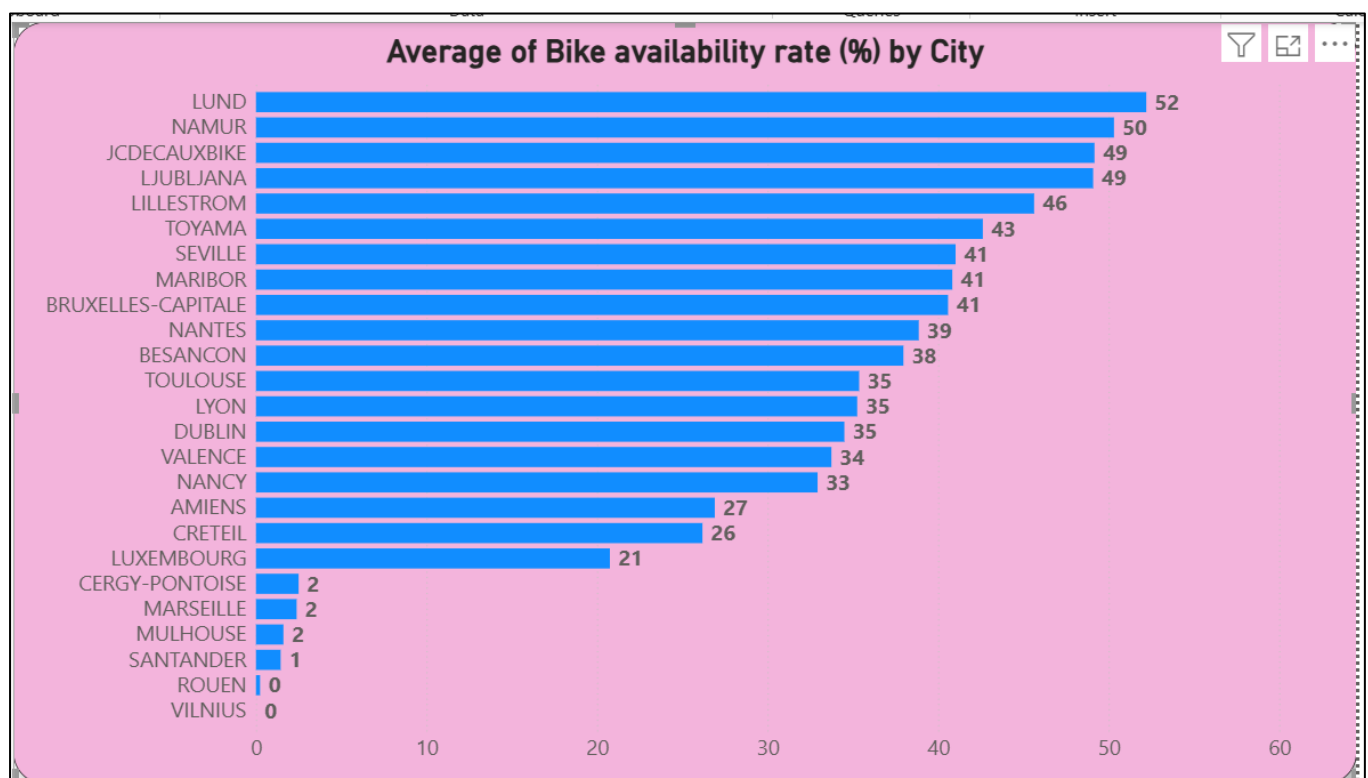


7. Analysis and Visualizations (Power BI)

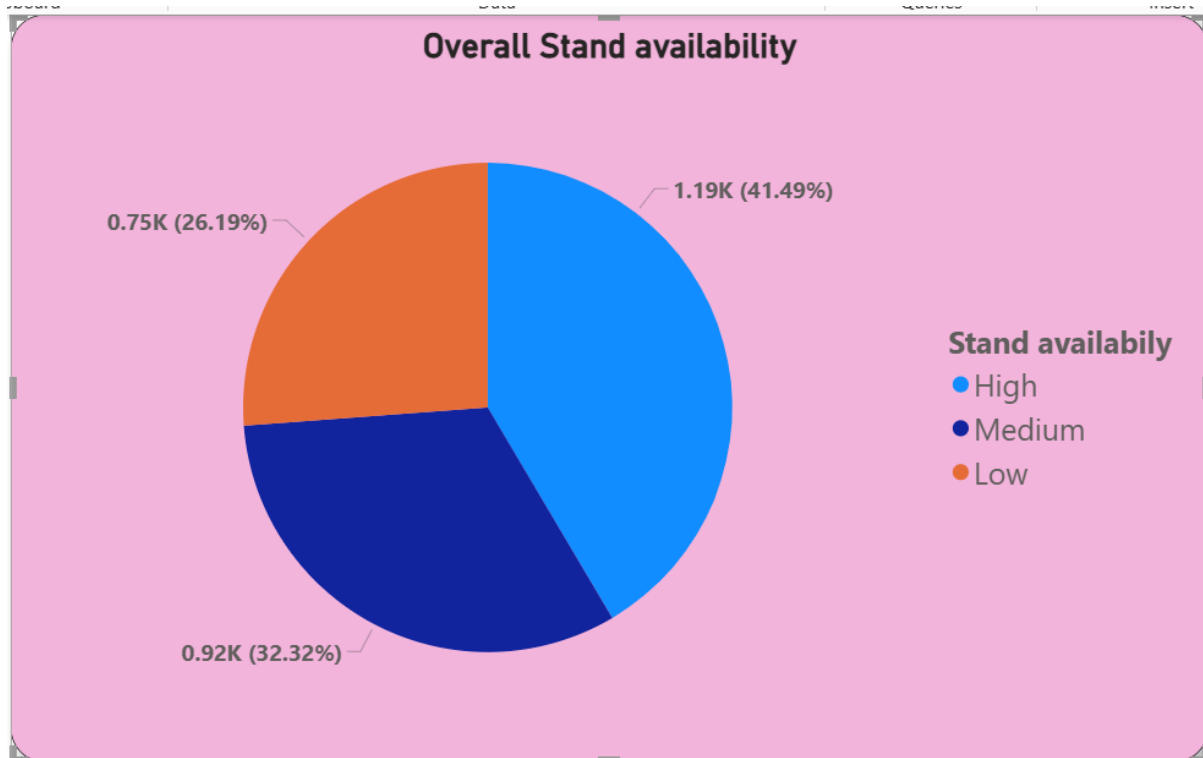


8. Insights & Conclusions

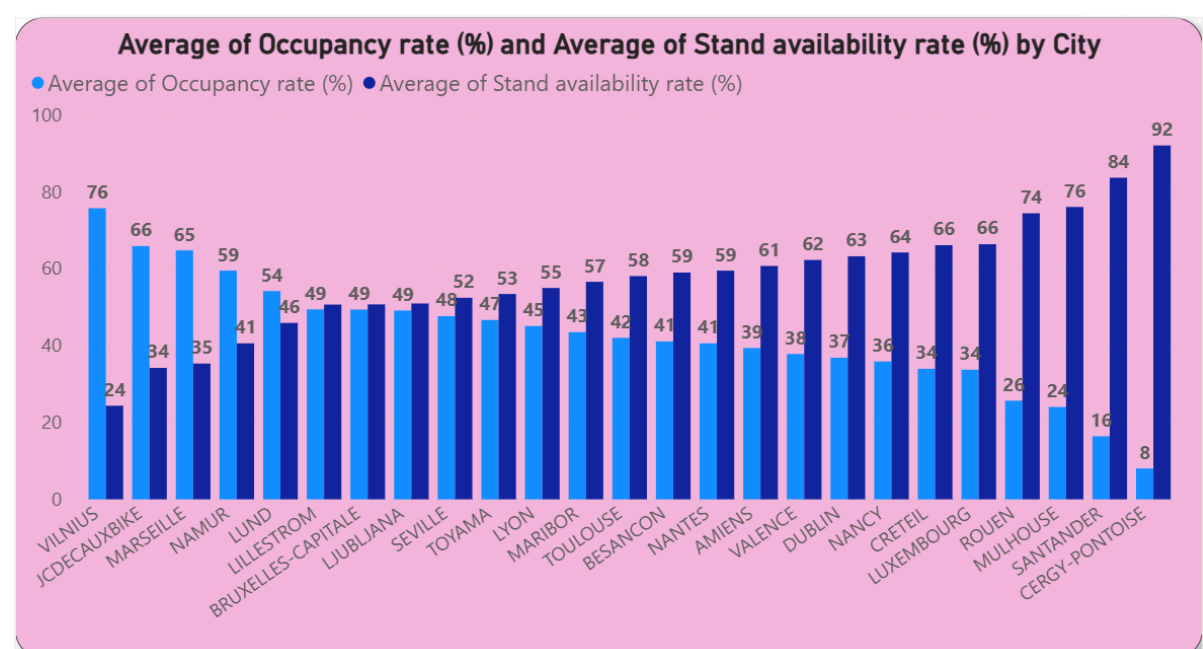
The percentages of average bike availability in each city are shown in the bar chart below. The ideal availability is between 40 to 60 percent considering a good balance between bike and stand availability. Out of 25, 9 cities perform well in terms of bike availability, 7 have medium performance, and the others are underperforming. The number of bikes has to be increased in those underperforming cities.



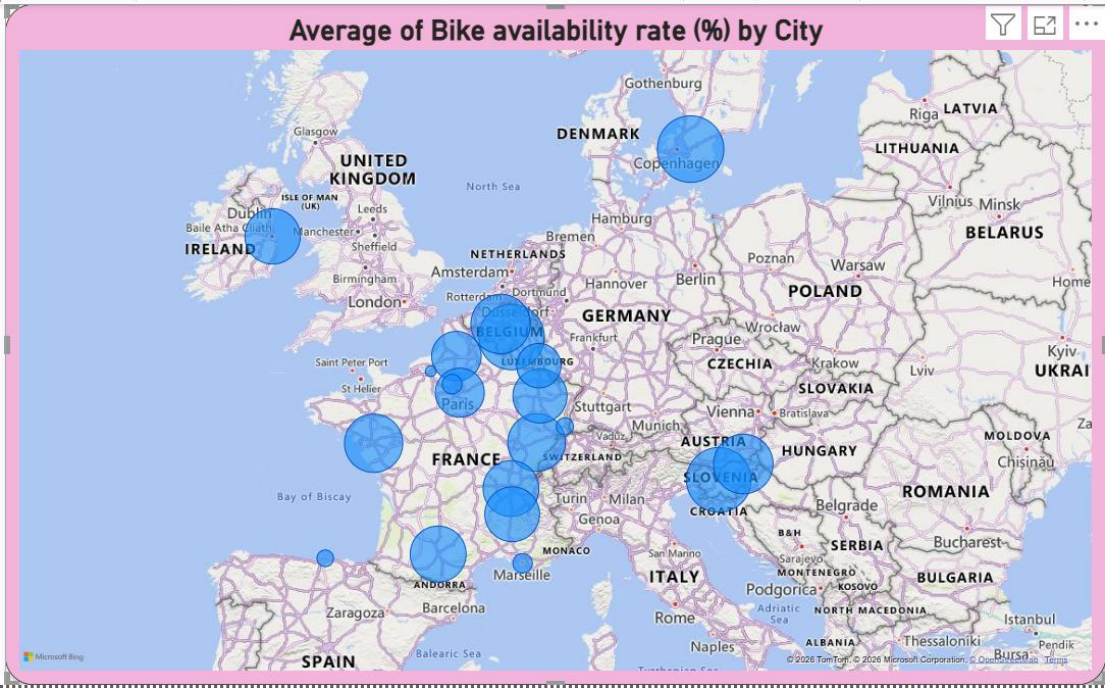
The availability of stands is categorized into High (more than 70%), Medium (30-70%), and Low (below 30%). This pie charts indicates that most of the cities have high availability and more than 30% cities have medium availability. However, a significant number of cities have low availability where focus is required on increasing the stands.



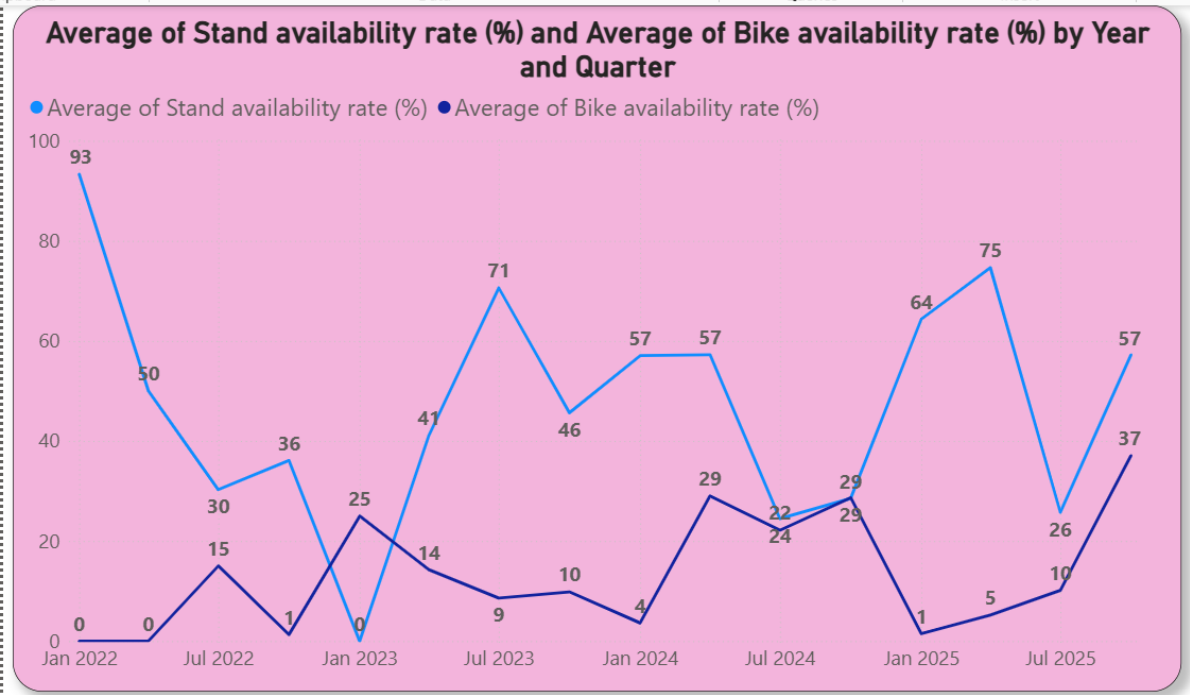
Ideally for a smooth operation, the balance between occupancy rate and stand availability rate is crucial. Individually these two categories should remain between 40-60% to maintain a good balance. Any number outside this range affects the balance significantly. The chart below shows this balance across cities.



The percentages of average bike availability in each city in shown in the map below. The ideal availability is between 40 to 60 percent considering a good balance between bike and stand availability. Out of 25, 9 cities perform well in terms of bike availability, 7 have medium performance, and the others are under performing. The number of bikes has to be increased in those underperforming cities.



The trends of average Stand availability and average Bike availability from 2022 to 2025 are shown below. The stand availability predominantly lies in the higher range whereas the bike availability lies in the lower range. This indicates the requirement to increase the number of bikes is more than that of the stands.



9. Conclusion

To achieve maximum operating efficiency, both the occupancy rate and the stand availability rate has to be balanced. To maintain these rates within the optimum range of 40-60% the number of stands is required to be increased in few cities and the number of bikes is required to be increased in few other cities as per the charts indicate.